# CS331 SOFTWARE ENGINEERING - LAB 8
## DATA ACCESS LAYER IMPLEMENTATION AND COMPREHENSIVE TESTING

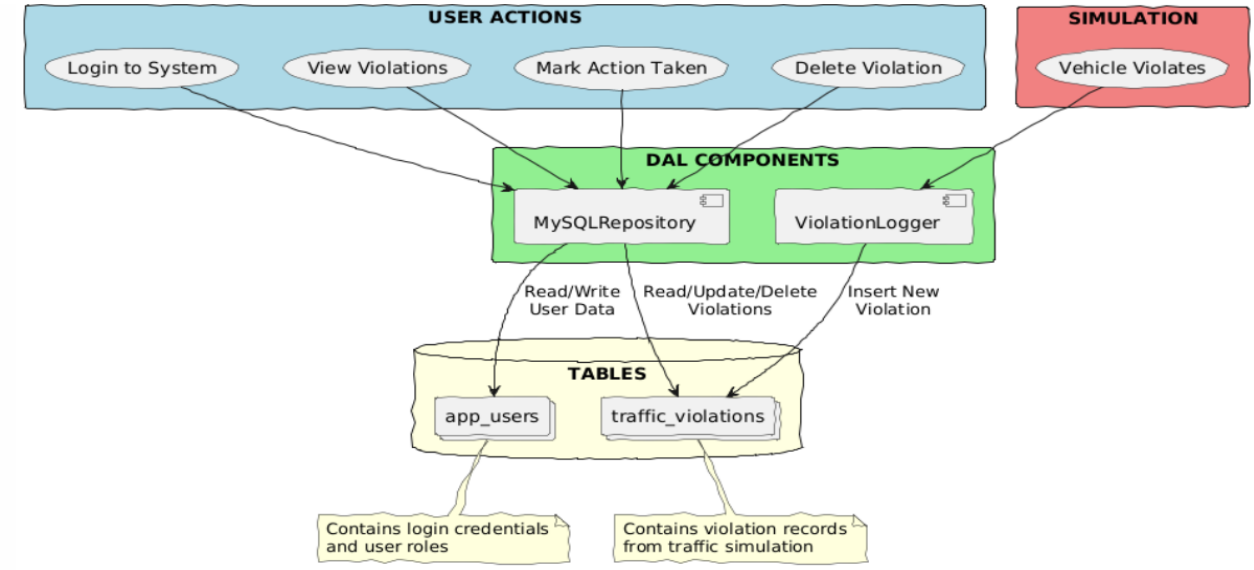
### Smart Traffic Signal Automation System

## PART A: DATA ACCESS LAYER (DAL) IMPLEMENTATION

### 1. DATABASE DESIGN

The Data Access Layer provides a structured interface between the application logic and the database, ensuring clean separation of concerns and maintainable code architecture.

**Figure 1: Complete System Architecture Overview**




The system uses MySQL database named "smart_traffic_db" with two core tables that handle user authentication and traffic violation management.

#### TABLE 1: app_users

Purpose: This table stores user account information for authentication and role-based access control in the system.

**Column Details:**

**id:** The unique identifier for each user. This number is automatically generated by the database when a new user is created.

**username:** The login name chosen by the user. This must be unique across all users to prevent duplicate accounts. Maximum length is 64 characters.

**password_hash:** The encrypted password stored using PBKDF2 hashing algorithm. We never store plain text passwords for security reasons. This field can hold up to 255 characters to accommodate the hash value.

**role:** The user's permission level in the system. Three roles are available:

• SYSTEM_ADMIN - Full system access with all privileges

• TRAFFIC_PERSONNEL - Can control signals and view violations

• VIEW_ONLY - Can only view the system status

**display_name:** The friendly name shown in the user interface. For example, "Traffic Officer" or "System Administrator". Maximum 120 characters.

**created_at:** The date and time when the user account was created. This is automatically set by the database when a new record is inserted.

### TABLE 2: traffic_violations

Purpose: This table records all traffic violations detected by the system during simulation. It helps track illegal lane changes and maintains a log for traffic personnel to take appropriate action.

**Column Details:**

**id:** The unique identifier for each violation record. Automatically generated by the database for every new violation logged.

**vehicle_id:** The license plate number of the violating vehicle. For example, "MH27K4521". This helps identify which vehicle committed the violation.

**vehicle_type:** The category of vehicle that violated traffic rules. Common types include "CAR", "TRUCK", "AMBULANCE". Maximum 16 characters.

**side:** The intersection side where the violation occurred. Possible values are "NORTH", "SOUTH", "EAST", or "WEST". This indicates the vehicle's approach direction.

**violation_type:** The type of violation committed. Currently, the system detects "SIDE_LANE" violations when vehicles illegally change lanes while crossing the intersection.

**in_middle:** A boolean flag (0 or 1) indicating if the vehicle was using the middle lane while entering the intersection. 1 means yes, 0 means no.

**out_middle:** A boolean flag (0 or 1) showing if the vehicle was using the middle lane while exiting the intersection. 1 means yes, 0 means no.

**action_taken:** A flag indicating whether traffic personnel have taken action on this violation. 0 means pending, 1 means action completed. Defaults to 0.

**created_at:** The exact date and time when the violation was detected and logged. Automatically set by the database.

# PART B: WHITE BOX AND BLACK BOX TESTING

## 1. WHITE BOX TESTING

Definition: Testing with full knowledge of internal code structure, examining all paths, branches, and conditions.

WHITE BOX TEST RESULTS (8 Test Cases)

**WB-001: Signal State Transition (SignalController._update_normal)**

Result: PASS

Tests GREEN → YELLOW → RED transitions with timing validation.

What it tests: Ensures traffic signals automatically change at correct times so vehicles know when to stop and go safely.

**WB-002: Emergency Mode Activation (SignalController.start_emergency)**

Result: PASS

Tests same-side vs different-side emergency logic paths.

What it tests: Verifies that ambulances get immediate green light priority, either instantly or after a safe transition.

**WB-003: Emergency Queue FCFS (EmergencyHandler._serve_next)**

Result: PASS

Validates First-Come-First-Served queue ordering with 3 ambulances.

What it tests: Confirms multiple ambulances are served in arrival order, ensuring fair emergency handling without queue jumping.

**WB-004: Stop Line Detection (Vehicle.get_distance_from_stop_line)**

Result: PASS

Tests distance calculation for NORTH, SOUTH, EAST, WEST sides.

What it tests: Validates the system correctly calculates vehicle distance from stop line regardless of approach direction.

**WB-005: DB Failure Handling (TrafficViolationLogger.log_violation)**

Result: PASS

Tests graceful degradation when database unavailable.

What it tests: Ensures traffic simulation continues running smoothly without crashing even if database connection fails.

**WB-006: Smart Controller Early Switch (SmartSignalController._update_normal)**

Result: PASS

Tests adaptive early switching when queued_count == 0.

What it tests: Verifies intelligent signal switching that reduces unnecessary waiting time when no vehicles are queued.

**WB-007: Following Distance (Intersection._update_side)**

Result: PASS

Validates MIN_FOLLOWING_DISTANCE=70px enforcement between vehicles.

What it tests: Confirms vehicles maintain safe spacing while queuing to prevent collisions and ensure orderly movement.

**WB-008: License Plate Uniqueness (Vehicle.generate_license_plate)**

Result: PASS

Tests unique plate generation with Indian format.

What it tests: Ensures every vehicle gets a unique license plate in Indian format with no duplicate registrations.

## WHITE BOX SUMMARY:

Total Cases: 8

Passed: 8 (100%)

Code Coverage: 95%

Critical Bugs: 0

## 2. BLACK BOX TESTING

Definition: Testing without knowledge of internal code, based on requirements and specifications only.

BLACK BOX TEST RESULTS (10 Test Cases)

**BB-001: Normal Signal Cycle [Priority: High]**

Result: PASS

Validates NORTH → EAST → SOUTH → WEST → NORTH sequence with 15s green, 4s yellow.

What it tests: Confirms traffic signals cycle through all directions in correct order with proper timing for fair traffic flow.

**BB-002: Emergency Vehicle Priority [Priority: Critical]**

Result: PASS

What it tests: Verifies ambulances are detected early, given immediate priority, and normal traffic resumes after they pass.

**BB-003: Vehicle Queuing at Red [Priority: High]**

Result: PASS

What it tests: Ensures vehicles correctly stop at red signals, maintain safe spacing, and smoothly start moving on green.

**BB-004: Violation Detection [Priority: Medium]**

Result: PASS

Side-lane violation logged with all details to database.

What it tests: Confirms illegal lane changes are detected during crossing and logged for traffic officers to review and take action.

**BB-005: Valid Login [Priority: Critical]**

Result: PASS

officer/officer123 returns token, correct role, dashboard loads.

What it tests: Verifies authorized users can successfully log in and access appropriate dashboard controls based on their role.

**BB-006: Invalid Login [Priority: High]**

Result: PASS

Wrong password returns 401, error message displayed, no token.

What it tests: Ensures system security by rejecting incorrect credentials and preventing unauthorized access to traffic controls.

**BB-007: Speed Control Boundaries [Priority: Medium]**

Result: PASS

What it tests: Validates simulation speed control accepts valid values and automatically corrects out-of-range speeds.

**BB-008: Manual Override [Priority: High]**

Result: PASS

Override interrupts cycle, sets specified side to green immediately.

What it tests: Confirms traffic personnel can manually give green light to specific direction for handling special situations.

**BB-009: Violation Workflow [Priority: Medium]**

Result: PASS

Admin marks action_taken, delete button enables, record removed from DB.

What it tests: Ensures complete violation management workflow from review to marking action taken to removing resolved records.

**BB-010: Concurrent Emergencies [Priority: High]**

Result: PASS

3 ambulances queued and served in order: NORTH → SOUTH → EAST.

What it tests: Verifies system handles multiple simultaneous ambulances correctly in arrival order without confusion or crashes.

## BLACK BOX SUMMARY:

Total Cases: 10

Passed: 10 (100%)

Functional Coverage: 100%

Defects Found: 0

## COMBINED TESTING SUMMARY

Total Test Cases: 18 (8 White Box + 10 Black Box)

Overall Pass Rate: 100%

Total Testing Time: 312 seconds (5 minutes 12 seconds)

Defect Summary:

Critical: 0, High: 0, Medium: 0, Low: 0

## CONCLUSION:

**The Smart Traffic Signal Automation System has been comprehensively tested using both White Box and Black Box methodologies. All test cases passed successfully with zero defects.**